Temperature probe

DETECT | RASPBERRY PI | DS18B20 | DIGITAL

What does it do?

A **DS18B20** measures temperature and sends a digital reading in degrees Celsius.

This probe has a sealed metal tip on a cable with three wires. Check the supplier's pinout; wire colours can vary.

Only the sealed probe goes in water. Keep the board and all wire connections dry.

Uses



Aquarium temperature alarm

DETECT Measure tank temperature → **DECIDE** Is it too cold? → **DO** Show a warning



Penguin bath alarm

DETECT Measure bath temperature → **DECIDE** Is it too warm? → **DO** Light a warning LED



VDD	DQ	GND	4.7 kΩ
Power from 3V3	Digital data	Ground	Resistor from DQ to 3V3

Raspberry Pi Foundation

Code

Raspberry Pi

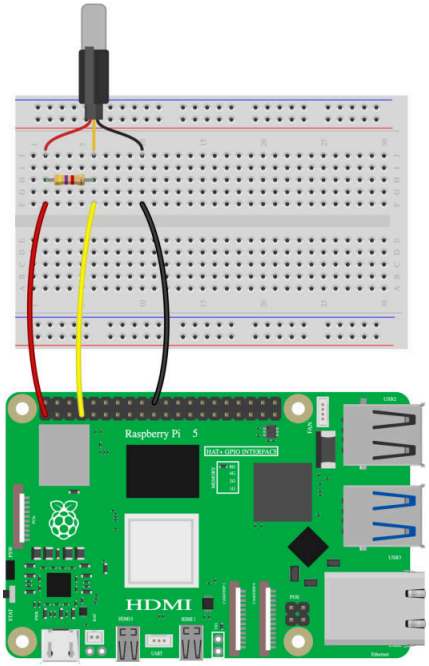
```
from w1thermsensor import W1ThermSensor

sensor = W1ThermSensor()
temperature = sensor.get_temperature()
print(temperature)
```

Setup: enable 1-Wire in `raspi-config`, reboot, then run `sudo apt install python3-w1thermsensor`.

Wiring

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VDD → 3V3 (pin 1); **DQ** → GPIO4 (pin 7); **GND** → GND (pin 20). **4.7 kΩ** between DQ and 3V3.